

Amazing

Fact and Science Fiction

stories

OCTOBER

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MAG 10

ESCAPE FROM ORBIT

by Poul Anderson

SF PROFILE
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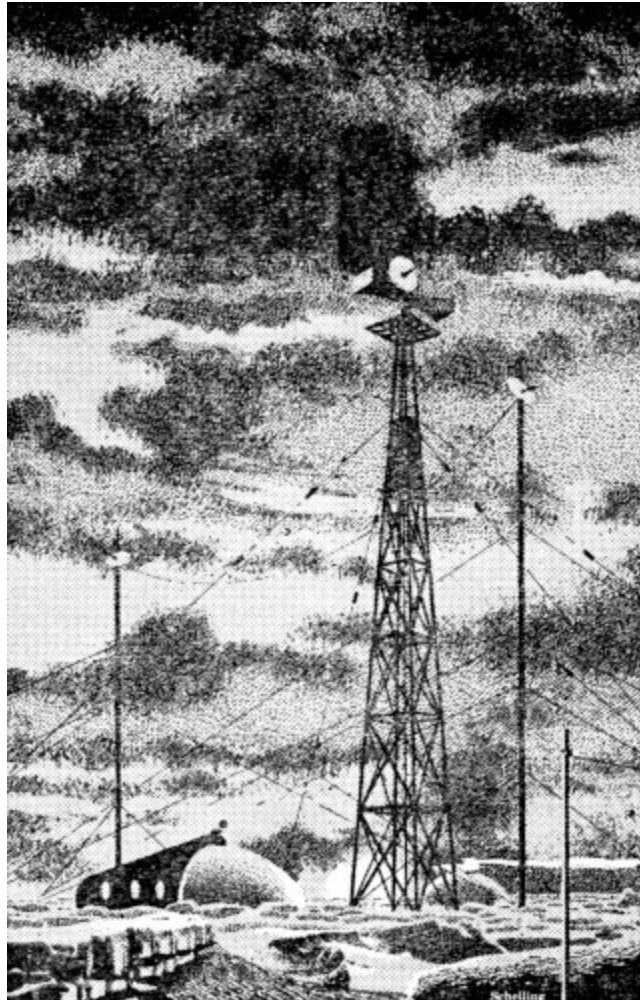
By BEN BOVA

Illustrated by SCHELLING

Astronomer Bova draws upon the facts of his field to weave a story that will grip your emotions and tantalize your mind—long after you have finished reading it.

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We had been at the South Pole a week. The outside thermometer read fifty degrees below zero, Fahrenheit. The winter was just beginning.

"What do you think we should transmit to McMurdo?" I asked Rizzo.

He put down his magazine and half-sat up in his bunk. For a moment there was silence, except for the nearly inaudible hum of the machinery that jammed our tiny dome, and the muffled shrieking of the ever-present wind, above us.

Rizzo looked at the semi-circle of control consoles, computers, and meteorological sensors with an expression of disgust that could be produced only by a drafted soldier.

"Tell 'em it's cold, it's gonna get colder, and we've both got appendicitis and need replacements immediately."

"Very clever," I said, and started touching the buttons that would automatically transmit the sensors' memory tapes.

Rizzo sagged back into his bunk. "Why?" He asked the curved ceiling of our cramped quarters. "Why me? Why here? What did I ever do to deserve spending the whole goddammed winter at the goddammed South Pole?"

"It's strictly impersonal," I assured him. "Some bright young meteorologist back in Washington has convinced the Pentagon that the South Pole is the key to the world's weather patterns. So here we are."

"It doesn't make sense," Rizzo continued, unhearing. His dark, broad-boned face was a picture of wronged humanity. "Everybody knows that when the missiles start flying, they'll be coming over the North Pole.... The goddammed Army is a hundred and eighty degrees off base."

"That's about normal for the Army, isn't it?" I was a drafted soldier, too.

Rizzo swung out of the bunk and paced across the dimly-lit room. It only took a half-dozen paces; the dome was small and most of it was devoted to machinery.

"Don't start acting like a caged lion," I warned. "It's going to be a long winter."

"Yeah, guess so." He sat down next to me at the radio console and pulled a pack of cigarets from his shirt pocket. He offered one to me, and we both smoked in silence for a minute or two.

"Got anything to read?"

I grinned. "Some microspool catalogues of stars."

"Stars?"

"I'm an astronomer ... at least, I was an astronomer, before the National Emergency was proclaimed."

Rizzo looked puzzled. "But I never heard of you."

"Why should you?"

"I'm an astronomer too."

"I thought you were an electronicist."

He pumped his head up and down. "Yeah ... at the radio astronomy observatory at Greenbelt. Project OZMA. Where do you work?"

"Lick Observatory ... with the 120-inch reflector."

"Oh ... an *optical* astronomer."

"Certainly."

"You're the first optical man I've met." He looked at me a trifle queerly.

I shrugged. "Well, we've been around a few millennia longer than you static-scanners."

"Yeah, guess so."

"I didn't realize that Project OZMA was still going on. Have you had any results yet?"

It was Rizzo's turn to shrug. "Nothing yet. The project has been shelved for the duration of the emergency, of course. If there's no war, and the dish doesn't get bombed out, we'll try again."

"Still listening to the same two stars?"

"Yeah ... Tau Ceti and Epsilon Eridani. They're the only two Sun-type stars within reasonable range that might have planets like Earth."

"And you expect to pick up radio signals from an intelligent race?"

"Hope to."

I flicked the ash off my cigaret. "You know, it always struck me as rather hopeless ... trying to find radio signals from intelligent creatures."

"Whattaya mean, hopeless?"

"Why should an intelligent race send radio signals out into interstellar space?" I asked. "Think of the power it requires, and the likelihood that it's all wasted effort, because there's no one within range to talk to."

"Well ... it's worth a try, isn't it ... if you think there could be intelligent creatures somewhere else ... on a planet of another star."

"Hmph. We're trying to find another intelligent race; are we transmitting radio signals?"

"No," he admitted. "Congress wouldn't vote the money for a transmitter that big."

"Exactly," I said. "We're listening, but not transmitting."

Rizzo wasn't discouraged. "Listen, the chances—just on statistical figuring alone—the chances are that there're millions of other solar systems with intelligent life. We've got to try contacting them! They might have knowledge that we don't have ... answers to questions that we can't solve yet...."

"I completely agree," I said. "But listening for radio signals is the wrong way to do it."

"Huh?"

"Radio broadcasting requires too much power to cover interstellar distances efficiently. We should be *looking* for signals, not listening for them."

"Looking?"

"Lasers," I said, pointing to the low-key lights over the consoles. "Optical lasers. Super-lamps shining out in the darkness of the void. Pump in a modest amount of electrical power, excite a few trillion atoms, and out comes a coherent, pencil-thin beam of light that can be seen for millions of miles."

"Millions of miles aren't lightyears," Rizzo muttered.

"We're rapidly approaching the point where we'll have lasers capable of lightyear ranges. I'm sure that some intelligent race somewhere in this galaxy has achieved the necessary technology to signal from star to star—by light beams."

"Then how come we haven't seen any?" Rizzo demanded.

"Perhaps we already have."

"What?"

"We've observed all sorts of variable stars—Cepheids, RR Lyrae's, T Tauri's. We assume that what we see are stars, pulsating and changing brightness for reasons that are natural, but unexplainable to us. Now, suppose what we are really viewing are laser beams, signalling from planets that circle stars too faint to be seen from Earth?"

In spite of himself, Rizzo looked intrigued.

"It would be fairly simple to examine the spectra of such light sources and determine whether they're natural stars or artificial laser beams."

"Have you tried it?"

I nodded.

"And?"

I hesitated long enough to make him hold his breath, waiting for my answer.

"No soap. Every variable star I've examined is a real star."

He let out his breath in a long, disgusted puff. "Ahhh, you were kidding all along. I thought so."

"Yes," I said. "I suppose I was."

Time dragged along in the weather dome. I had managed to smuggle a small portable telescope along with me, and tried to make observations whenever possible. But the weather was usually too poor. Rizzo, almost in desperation for something to do, started to build an electronic image-amplifier for me.

Our one link with the rest of the world was our weekly radio message from McMurdo. The times for the messages were randomly scrambled, so that the chances of their being intercepted or jammed were lessened. And we were ordered to maintain strict radio silence.

As the weeks sloughed on, we learned that one of our manned satellites had been boarded by the Reds at gunpoint. Our space-crews had put two Red automated spy-satellites out of commission. Shots had been exchanged on an ice-island in the Arctic. And six different nations were testing nuclear bombs.

We didn't get any mail of course. Our letters would be waiting for us at McMurdo when we were relieved. I thought about Gloria and our two children quite a bit, and tried not to think about the blast and fallout patterns in the San Francisco area, where they were.

"My wife hounded me until I spent pretty nearly every damned cent I had on a shelter, under the house," Rizzo told me. "Damned shelter is fancier than the house. She's the social leader of the disaster set. If we don't have a war, she's gonna feel damned silly."

I said nothing.

The weather cleared and steadied for a while (days and nights were indistinguishable during the long Antarctic winter) and I split my time evenly between monitoring the meteorological sensors and observing the stars. The snow had covered the dome completely, of course, but our "snorkel" burrowed through it and out into the air.

"This dome's just like a submarine, only we're submerged in snow instead of water," Rizzo observed. "I just hope we don't sink to the bottom."

"The calculations show that we'll be all right."

He made a sour face. "Calculations proved that airplanes would never get off the ground."

The storms closed in again, but by the time they cleared once more, Rizzo had completed the image-amplifier for me. Now, with the tiny telescope I had, I could see almost as far as a professional instrument would allow. I could even lie comfortably in my bunk, watch the amplifier's viewscreen, and control the entire set-up remotely.

Then it happened.

At first it was simply a curiosity. An oddity.

I happened to be studying a Cepheid variable star—one of the huge, very bright stars that pulsate so regularly that you can set your watch by them. It had attracted my attention because it seemed to be unusually close for a Cepheid—only 700 lightyears away. The distance could be easily gauged by timing the star's pulsations.^[1]

I talked Rizzo into helping me set up a spectrometer. We scavenged shamelessly from the dome's spare parts bin and finally produced an instrument that would break up the light of the star into its component wavelengths, and thereby tell us much about the star's chemical composition and surface temperature.

At first I didn't believe what I saw.

The star's spectrum—a broad rainbow of colors—was criss-crossed with narrow dark lines. That was all right. They're called absorption lines; the Sun has thousands of them in its spectrum. But one line—*one*—was an insolently bright emission line. All the laws of physics and chemistry said it couldn't be there.

But it was.

We photographed the star dozens of times. We checked our instruments ceaselessly. I spent hours scanning the star's "official" spectrum in the microspool reader. The bright emission line was not on the catalogue spectrum. There was nothing wrong with our instruments.

Yet the bright line showed up. It was real.

"I don't understand it," I admitted. "I've seen stars with bright emission spectra before, but a single bright line in an absorption spectrum! It's unheard-of. One single wavelength ... one particular type of atom at one precise energy-level ... why? Why is it emitting energy when the other wavelengths aren't?"

Rizzo was sitting on his bunk, puffing a cigaret. He blew a cloud of smoke at the low ceiling. "Maybe it's one of those laser signals you were telling me about a couple weeks ago."

I scowled at him. "Come on, now. I'm serious. This thing has me puzzled."

"Now wait a minute ... you're the one who said radio astronomers were straining their ears for nothing. You're the one who said we ought to be

looking. So look!" He was enjoying his revenge.

I shook my head, and turned back to the meteorological equipment.

But Rizzo wouldn't let up. "Suppose there's an intelligent race living on a planet near a Cepheid variable star. They figure that any other intelligent creatures would have astronomers who'd be curious about their star, right? So they send out a laser signal that matches the star's pulsations. When you look at the star, you see their signal. What's more logical?"

"All right," I groused. "You've had your joke...."

"Tell you what," he insisted. "Let's put that one wavelength into an oscilloscope and see if a definite signal comes out. Maybe it'll spell out 'Take me to your leader' or something."

I ignored him and turned my attention to Army business. The meteorological equipment was functioning perfectly, but our orders read that one of us had to check it every twelve hours. So I checked and tried to keep my eyes from wandering as Rizzo tinkered with a photocell and oscilloscope.

"There we are," he said, at length. "Now let's see what they're telling us."

In spite of myself I looked up at the face of the oscilloscope. A steady, gradually sloping greenish line was traced across the screen.

"No message," I said.

Rizzo shrugged elaborately.

"If you leave the 'scope on for two days, you'll find that the line makes a full swing from peak to null," I informed him. "The star pulsates every two days, bright to dim."

"Let's turn up the gain," he said, and he flicked a few knobs on the front of the 'scope.

The line didn't change at all.

"What's the sweep speed?" I asked.

"One nanosecond per centimeter." That meant that each centimeter-wide square on the screen's face represented one billionth of a second. There are as many nanoseconds in one second as there are seconds in thirty-two years.

"Well, if you don't get a signal at that sensitivity, there just isn't any signal there," I said.

Rizzo nodded. He seemed slightly disappointed that his joke was at an end. I turned back to the meteorological instruments, but I couldn't concentrate on them. Somehow I felt disappointed, too. Subconsciously, I suppose, I had been hoping that Rizzo actually would detect a signal from the star. *Fool!* I told myself. But what could explain that bright emission line? I glanced up at the oscilloscope again.

And suddenly the smooth steady line broke into a jagged series of millions of peaks and nulls!

I stared at it.

Rizzo was back on his bunk again, reading one of his magazines. I tried to call him, but the words froze in my throat. Without taking my eyes from the flickering 'scope, I reached out and touched his arm.

He looked up.

"Holy Mother of God," Rizzo whispered.

For a long time we stared silently at the fluttering line dancing across the oscilloscope screen, bathing our tiny dome in its weird greenish light. It was eerily fascinating, hypnotic. The line never stood still; it jabbered and stuttered, a series of millions of little peaks and nulls, changing almost too fast for the eye to follow, up and down, calling to us, speaking to us, up, down, never still, never quiet, constantly flickering its unknown message to us.



The line never stood still; millions of little peaks and nulls calling to us, speaking to us, never still, never quiet, constantly flickering its unknown message to us.

"Can it be ... people?" Rizzo wondered. His face, bathed in the greenish light, was suddenly furrowed, withered, ancient: a mixture of disbelief and fear.

"What else could it be?" I heard my own voice answer. "There's no other explanation possible."

We sat mutely for God knows how long.

Finally Rizzo asked, "What do we do now?"

The question broke our entranced mood. What do we do? What action do we take? We're thinking men, and we've been contacted by other creatures that can think, reason, send a signal across seven hundred lightyears of space. So don't just sit there in stupified awe. Use your brain, prove that you're worthy of the tag *sapiens*.

"We decode the message," I announced. Then, as an after-thought, "But don't ask me how."

We should have called McMurdo, or Washington. Or perhaps we should have attempted to get a message through to the United Nations. But we never even thought of it. This was our problem. Perhaps it was the sheer isolation of our dome that kept us from thinking about the rest of the world. Perhaps it was sheer luck.

"If they're using lasers," Rizzo reasoned, "they must have a technology something like ours."

"Must *have had*," I corrected. "That message is seven hundred years old, remember. They were playing with lasers when King John was signing the Magna Charta and Genghis Khan owned most of Asia. Lord knows what they have now."

Rizzo blanched and reached for another cigaret.

I turned back to the oscilloscope. The signal was still flashing across its face.

"They're sending out a signal," I mused, "probably at random. Just beaming it out into space, hoping that someone, somewhere will pick it up. It must be in some form of code ... but a code that they feel can be easily cracked by anyone with enough intelligence to realize that there's a message there."

"Sort of an interstellar Morse code."

I shook my head. "Morse code depends on both sides knowing the code. There's no key."

"Cryptographers crack codes."

"Sure. If they know what language is being used. We don't know the language, we don't know the alphabet, the thought processes ... nothing."

"But it's a code that can be cracked easily," Rizzo muttered.

"Yes," I agreed. "Now what the hell kind of a code can they assume will be known to another race that they've never seen?"

Rizzo leaned back on his bunk and his face was lost in shadows.

"An interstellar code," I rambled on. "Some form of presenting information that would be known to almost any race intelligent enough to understand lasers...."

"Binary!" Rizzo snapped, sitting up on the bunk.

"What?"

"Binary code. To send a signal like this, they've gotta be able to write a message in units that're only a billionth of a second long. That takes computers. Right? Well, if they have computers, they must figure that we have computers. Digital computers run on binary code. Off or on ... go or no-go. It's simple. I'll bet we can slap that signal on a tape and run it through our computer here."

"To assume that they use computers exactly like ours...."

"Maybe the computers are completely different," Rizzo said excitedly, "but the binary code is basic to them all. I'll bet on that! And this computer we've got here—this transistorized baby—she can handle more information than the whole Army could feed into her. I'll bet nothing has been developed anywhere that's better for handling simple one-plus-one types of operations."

I shrugged. "All right. It's worth a trial."

It took Rizzo a few hours to get everything properly set up. I did some arithmetic while he worked. If the message was in binary code, that meant that every cycle of the signal—every flick of the dancing line on our screen—carried a bit of information. The signal's wavelength was 5000

Angstroms; there are a hundred million Angstrom units to the centimeter; figuring the speed of light ... the signal could carry, in theory at least, something like 600 trillion bits of information *per second*.

I told Rizzo.

"Yeah, I know. I've been going over the same numbers in my head." He set a few switches on the computer control board. "Now let's see how many of the 600 trillion we can pick up." He sat down before the board and pressed a series of buttons.

We watched, hardly breathing, as the computer's spools began spinning and the indicator lights flashed across the control board. Within a few minutes, the printer chugged to life.

Rizzo swivelled his chair over to the printer and held up the unrolling sheet in a trembling hand.

Numbers. Six-digit numbers. Completely meaningless.

"Gibberish," Rizzo snapped.

It was peculiar. I felt relieved and disappointed at the same time.

"Something's screwy," Rizzo said. "Maybe I fouled up the circuits...."

"I don't think so," I answered. "After all, what did you expect out of the computer? Shakespearean poetry?"

"No, but I expected numbers that would make some sense. One and one, maybe. Something that means something. This stuff is nowhere."

Our nerves must have really been wound tight, because before we knew it we were in the middle of a nasty argument—and it was over nothing, really. But in the middle of it:

"Hey, look," Rizzo shouted, pointing to the oscilloscope.

The message had stopped. The 'scope showed only the calm, steady line of the star's basic two-day-long pulsation.

It suddenly occurred to us that we hadn't slept for more than 36 hours, and we were both exhausted. We forgot the senseless argument. The message was ended. Perhaps there would be another; perhaps not. We had the telescope, spectrometer, photocell, oscilloscope, and computer set to record

automatically. We collapsed into our bunks. I suppose I should have had monumental dreams. I didn't. I slept like a dead man.

When we woke up, the oscilloscope trace was still quiet.

"Y'know," Rizzo muttered, "it might just be a fluke ... I mean, maybe the signals don't mean a damned thing. The computer is probably translating nonsense into numbers just because it's built to print out numbers and nothing else."

"Not likely," I said. "There are too many coincidences to be explained. We're receiving a message, I'm certain of it. Now we've got to crack the code."

As if to reinforce my words, the oscilloscope trace suddenly erupted into the same flickering pattern. The message was being sent again.

We went through two weeks of it. The message would run through for seven hours, then stop for seven. We transcribed it on tape 48 times and ran it through the computer constantly. Always the same result—six-digit numbers; millions of them. There were six different seven-hour-long messages, being repeated one after the other, constantly.

We forgot the meteorological equipment. We ignored the weekly messages from McMurdo. The rest of the world became a meaningless fiction to us. There was nothing but the confounded, tantalizing, infuriating, enthralling message. The National Emergency, the bomb tests, families, duties—all transcended, all forgotten. We ate when we thought of it and slept when we couldn't keep our eyes open any longer. The message. What was it? What was the key to unlock its meaning?

"It's got to be something universal," I told Rizzo. "Something universal ... in the widest sense of the term."

He looked up from his desk, which was wedged in between the end of his bunk and the curving dome wall. The desk was littered with printout sheets from the computer, each one of them part of the message.

"You've only said that a half-million times in the past couple weeks. What the hell is universal? If you can figure that out, you're damned good."

What is universal? I wondered. *You're an astronomer. You look out at the universe. What do you see?* I thought about it. *What do I see?* Stars, gas, dust clouds, planets ... *what's universal about them? What do they all have that....*

"Atoms!" I blurted.

Rizzo cocked a weary eye at me. "Atoms?"

"Atoms. Elements. Look..." I grabbed up a fistful of the sheets and thumbed through them. "Look ... each message starts with a list of numbers. Then there's a long blank to separate the opening list from the rest of the message. See? Every time, the same length list."

"So?"

"The periodic table of the elements!" I shouted into his ear. "That's the key!"

Rizzo shook his head. "I thought of that two days ago. No soap. In the first place, the list that starts each message isn't always the same. It's the same length, all right, but the numbers change. In the second place, it always begins with 100000. I looked up the atomic weight of hydrogen—it's 1.008 something."

That stopped me for a moment. But then something clicked into place in my mind.

"Why is the hydrogen weight 1.008?" Before Rizzo could answer, I went on, "For two reasons. The system we use arbitrarily rates oxygen as 16-even. Right? All the other weights are calculated from oxygen's. And we also give the average weight of an element, counting all its isotopes. Our weight for hydrogen also includes an adjustment for tiny amounts of deuterium and tritium. Right? Well, suppose *they* have a system that rates hydrogen as a flat one: 1.00000. Doesn't *that* make sense?"

"You're getting punchy," Rizzo grumbled. "What about the isotopes? How can they expect us to handle decimal points if they don't tell us about them ... mental telepathy? What about...."

"Stop arguing and start calculating," I snapped. "Change that list of numbers to agree with our periodic table. Change 1.00000 to 1.008-whatever-it-is and tackle the next few elements. The decimals shouldn't be so hard to figure out."

Rizzo grumbled to himself, but started working out the calculations. I stepped over to the dome's microspool library and found an elementary physics text. Within a few minutes, Rizzo had some numbers and I had the periodic table focused on the microspool reading machine.

"Nothing," Rizzo said, leaning over my shoulder and looking at the screen. "They don't match at all."

"Try another list. They're not all the same."

He shrugged and returned to his desk. After a while he called out, "their second number is 3.97123; it works out to 4.003-something."

It checked! "Good. That's helium. What about the next one, lithium?"

"That's 6.940."

"Right!"

Rizzo went to work furiously after that. I pushed a chair to the desk and began working up from the end of the list. It all checked out, from hydrogen to a few elements beyond the artificial ones that had been created in the laboratories here on Earth.

"That's it," I said. "That's the key. That's our Rosetta Stone ... the periodic table."

Rizzo stared at the scribbled numbers and jumble of papers. "I bet I know what the other lists are ... the ones that don't make sense."

"Oh?"

"There are other ways to identify the elements ... vibration resonances, quantum wavelengths ... somebody named Lewis came out a couple years ago with a Quantum Periodic Table...."

"They're covering all the possibilities. There are messages for many different levels of understanding. We just decoded the simplest one."

"Yeah."

I noticed that as he spoke, Rizzo's hand—still tightly clutching the pencil—was trembling and white with tension.

"Well?"

Rizzo licked his lips. "Let's get to work."

We were like two men possessed. Eating, sleeping, even talking was ignored completely as we waded through the hundreds of sheets of paper. We could decode only a small percentage of them, but they still represented many hours of communication. The sheets that we couldn't decode, we suspected, were repetitions of the same message that we were working on.

We lost all concept of time. We must have slept, more than once, but I simply don't remember. All I can recall is thousands of numbers, row upon row, sheet after sheet of numbers ... and my pencil scratching symbols of the various chemical elements over them until my hand was so cramped I could no longer open the fingers.

The message consisted of a long series of formulas; that much was certain. But, without punctuation, with no knowledge of the symbols that denote even such simple things as "plus" or "equals" or "yields," it took us more weeks of hard work to unravel the sense of each equation. And even then, there was more to the message than met the eye:

"Just what the hell are they driving at?" Rizzo wondered aloud. His face had changed: it was thinner, hollow-eyed, weary, covered with a scraggly beard.

"Then you think there's a meaning behind all these equations, too?"

He nodded. "It's a message, not just a contact. They're going to an awful lot of trouble to beam out this message, and they're repeating it every seven hours. They haven't added anything new in the weeks we've been watching."

"I wonder how many years or centuries they've been sending out this message, waiting for someone to pick it up, looking for someone to answer them."

"Maybe we should call Washington...."

"No!"

Rizzo grinned. "Afraid of breaking radio silence?"

"Hell no. I just want to wait until we're relieved, so we can make this announcement in person. I'm not going to let some old wheezer in Washington get credit for this.... Besides, I want to know just what they're trying to tell us."

It was agonizing, painstaking work. Most of the formulas meant nothing to either one of us. We had to ransack the dome's meager library of microspools to piece them together. They started simply enough—basic chemical combinations: carbon and two oxygens yield CO_2 ; two hydrogens and oxygen give water. A primer ... not of words, but of equations.

The equations became steadily longer and more complex. Then, abruptly, they simplified, only to begin a new deepening, simplify again, and finally become very complicated just at the end. The last few lines were obviously repetitious.

Gradually, their meaning became clear to us.

The first set of equations started off with simple, naturally-occurring energy yielding formulas. The oxidation of cellulose (we found the formula for that in an organic chemistry text left behind by one of the dome's previous occupants), which probably referred to the burning of plants and vegetation. A string of formulas that had groupings in them that I dimly recognized as amino acids—no doubt something to do with digesting food. There were many others, including a few that Rizzo claimed had the expression for chlorophyll in them.

"Naturally-occurring, energy-yielding reactions," Rizzo summarized. "They're probably trying to describe the biological set-up on their planet."

It seemed an inspired guess.

The second set of equations again began with simple formulas. The cellulose-burning reaction appeared again, but this time it was followed by equations dealing with the oxidation of hydrocarbons: coal and oil burning? A long series of equations that bore repeatedly the symbols for many different metals came up next, followed by more on hydrocarbons, and then a string of formulas that we couldn't decipher at all.

This time it was my guess: "These look like energy-yielding reactions, too. At least in the beginning. But they don't seem to be naturally occurring types. Then comes a long story about metals. They're trying to tell us the history of their technological development—burning wood, coal and eventually oil; smelting metals ... they're showing us how they developed their technology."

The final set of equations began with an ominous simplicity: a short series of very brief symbols that had the net result of four hydrogen atoms building into a helium atom. Nuclear fusion.

"That's the proton-proton reaction," I explained to Rizzo. "The type of fusion that goes on in the Sun."

The next series of equations spelled out the more complex carbon-nitrogen cycle of nuclear fusion, which was probably the primary energy source of their own Cepheid variable star. Then came a long series of equations that we couldn't decode in detail, but the symbols for uranium and plutonium, and some of the heavier elements, kept cropping up.

Then came one line that told us the whole story: the lithium-hydride equation—nuclear fusion bombs.

The equations went on to more complex reactions, formulas that no man on Earth had ever seen before. They were showing us the summation of their knowledge, and they had obviously been dealing with nuclear energies for much longer than we have on Earth.

But interspersed among the new equations, they repeated a set of formulas that always began with the lithium-hydride fusion reaction. The message ended in a way that wrenched my stomach: the fusion bomb reaction and its cohorts were repeated ten straight times.

I'm not sure of what day it was on the calendar, but the clock on the master control console said it was well past eleven.

Rizzo rubbed a weary hand across his eyes. "Well, what do you think?"

"It's pretty obvious," I said. "They have the bombs. They've had them for quite some time. They must have a lot of other weapons, too—more ... advanced. They're trying to tell us their history with the equations. First they depended on natural sources of energy, plants and animals; then they developed artificial energy sources and built up a technology; finally they discovered nuclear energy."

"How long do you think they've had the bombs?"

"Hard to tell. A generation ... a century. What difference does it make? They have them. They probably thought, at first, that they could learn to live with them ... but imagine what it must be like to have those weapons at your fingertips ... for a century. Forever. Now they're so scared of them that they're beaming their whole history out into space, looking for someone to tell them how to live with the bombs, how to avoid using them."

"You could be wrong," Rizzo said. "They could be boasting about their arsenal."

"Why? For what reason? No ... the way they keep repeating those last equations. They're pleading for help."

Rizzo turned to the oscilloscope. It was flickering again.

"Think it's the same thing?"

"No doubt. You're taping it anyway, aren't you?"

"Yeah, sure. Automatically."

Suddenly, in mid-flight, the signal winked off. The pulsations didn't simply smooth out into a steady line, as they had before. The screen simply went dead.

"That's funny," Rizzo said, puzzled. He checked the oscilloscope. "Nothing wrong here. Something must've happened to the telescope."

Suddenly I knew what had happened. "Take the spectrometer off and turn on the image-amplifier," I told him.

I knew what we would see. I knew why the oscilloscope beam had suddenly gone off scale. And the knowledge was making me sick.

Rizzo removed the spectrometer set-up and flicked the switch that energized the image-amplifier's viewscreen.

"Holy God!"

The dome was flooded with light. The star had exploded.

"They had the bombs all right," I heard myself saying. "And they couldn't prevent themselves from using them. And they had a lot more, too. Enough to push their star past its natural limits."

Rizzo's face was etched in the harsh light.

"I've gotta get out of here," he muttered, looking all around the cramped dome. "I've gotta get back to my wife and find someplace where it's safe...."

"Someplace?" I asked, staring at the screen. "Where?"

THE END

[1] Astronomers have been able, since about 1910, to estimate the distances of Cepheid variable stars by timing their pulsations. The length of this type of star's pulsation is a true measure of its intrinsic brightness. Comparing the star's actual brightness to its apparent brightness, as seen from Earth, gives a good value for the star's distance.

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